

HAFIZ HILAL AHMAD

MS Structural Engineering Candidate · Civil Engineer (PEC Reg. No. Civil/59341)

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RESEARCH STATEMENT

Structural engineer with hands-on experimental research experience in the seismic behaviour of confined masonry and an industry background in large-scale infrastructure construction. Currently completing an MS in Structural Engineering at NUST Islamabad, with thesis work on the dynamic performance of a confined asymmetrical masonry building investigated through shake-table testing and finite-element analysis. Strongly motivated to pursue doctoral research at the ROSE PhD program, with interests in earthquake engineering, performance-based seismic assessment of confined and unreinforced masonry structures, experimental and numerical modelling of masonry behaviour, and seismic vulnerability and risk assessment of existing buildings in low- and middle-income regions.

EDUCATION

MS in Structural Engineering · National University of Sciences and Technology (NUST), Islamabad

Oct 2023 – Present (expected completion: September 2026) · CGPA: 3.38 / 4.00

- MS thesis: “Dynamic Performance of a Confined Asymmetrical Masonry Building — Shake-Table Testing and Finite-Element Analysis.”
- Relevant coursework: Advanced Structural Dynamics, Fire Structural Engineering, Finite Element Analysis, Advanced Concrete Structures, Advanced Structural Mechanics.

BS in Civil Engineering · International Islamic University, Islamabad (IIUI)

2016 – 2020 · CGPA: 3.07 / 4.00

- Final-year project: Structural fire rating of an office building in Islamabad (structural fire engineering).
- Foundational coursework in Structural Analysis, Reinforced Concrete Design, Steel Design, Geotechnical Engineering, and Construction Management.

Higher Secondary School Certificate (FSc – Pre-Engineering) · HITEC College, Taxila Cantt.

Completed 2015 · Grade: A

Secondary School Certificate (SSC – Science) · HITEC College, Taxila Cantt.

2010 – 2012 · Grade: A1

RESEARCH EXPERIENCE

Laboratory Engineer · Structural Dynamics Lab, MCE NUST Islamabad (Risalpur Campus)

Nov 2025 - Present

Graduate Researcher · Department of Structural Engineering, NIT NUST Islamabad (Risalpur Campus)

Oct 2023 – Oct 2025

- Investigating the dynamic seismic performance of a confined asymmetrical masonry building through scaled shake-table testing, with the aim of characterizing its global response, torsional behaviour, damage progression, and failure mechanisms under recorded ground motions.
- Developing parallel finite-element (FEM) models calibrated against the shake-table results, in order to validate numerical predictions, quantify modelling uncertainties, and extend the experimental findings through parametric studies.
- Hands-on experience with experimental setup design, instrumentation (accelerometers, LVDTs, strain gauges), data acquisition, signal processing, and post-test damage characterization.
- Developing competence in nonlinear time-history analysis and macro/micro-modelling of masonry in ABAQUS, with a focus on confined-masonry constitutive behaviour and asymmetric in-plan response.

Undergraduate Research Assistant – Structural Fire Engineering · IIU Islamabad

2019 – 2020

- Investigated structural fire rating of an office building, including assessment of fire load and probabilistic temperature exposure of structural members.
- Resulting work published in a peer-reviewed journal (see Publications).

PUBLICATIONS

1. M. Noman, **H. H. Ahmad**, A. Hamza, and M. Yaqub, “Assessment of Fire Load and Probabilistic Temperature for Office Buildings in Pakistan,” *Civil Engineering Infrastructures Journal*, pp. 1–27, 2022. DOI: 10.22059/CEIJ.2022.341199.1827.

RESEARCH INTERESTS

- Earthquake engineering of masonry structures (confined, unreinforced, and retrofitted masonry).
- Shake-table testing and dynamic identification of structural systems.
- Performance-based seismic assessment and design of existing buildings.
- Nonlinear finite-element modelling of masonry under seismic loading, including torsional/asymmetric response.
- Seismic vulnerability and risk assessment of buildings in developing regions.

PROFESSIONAL EXPERIENCE

Laboratory Engineer · Structural Dynamics Lab, MCE NUST Islamabad (Risalpur Campus)

Nov 2025 - Present

Assistant Project Manager · Frontier Works Organization (FWO), Pakistan

April 2022 – October 2023 (1 year, 7 months)

- **Rawalpindi Ring Road Project (RRR), Phase-I** — 38.3 km from Banth (N-5) to Thalian Interchange (M-2). Supervised earthwork, structural works, and on-site planning under the 751 CTE FWO command.
- **Kurram Tangi Multipurpose Dam Project, Phase-I** (764 CTE FWO) — supervised construction of tunnels, canals, and water-retaining structures for an 18.4 MW hydropower and 16,300-acre irrigation scheme.
- Coordinated multidisciplinary site teams, monitored compliance with construction standards, and contributed to project planning and progress reporting.

Site Engineer · Alzeb Burki Construction Company (SMC – Pvt) Ltd

June 2021 – March 2022 (10 months)

- Sub-contractor to FWO (763 CTE) on the Thall–Mirali Road (54 km) project in North Waziristan.
- Ensured timely execution of scheduled activities, quality control on site, and adherence to engineering specifications. Supported project-management functions including progress monitoring and reporting.

Construction Inspector (Trainee) · EA Consulting (Pvt) Ltd

July 2019 – August 2019 (2 months)

- Site inspector during construction of a high-rise residential development in G-13, Islamabad.

TECHNICAL SKILLS

Structural analysis & FEM: CSI ETABS, CSI SAFE, Abaqus, SAP2000 (basic).

BIM / CAD: AutoCAD, Autodesk Revit.

Programming & data: Python (NumPy, pandas, Matplotlib), Microsoft Excel (advanced).

Productivity: Microsoft Word, PowerPoint; LaTeX (basic).

Languages: English (professional working proficiency), Urdu (native), Pashto (native).

LICENCES, CERTIFICATIONS & MEMBERSHIPS

- Registered Engineer — Pakistan Engineering Council (PEC), Reg. No. Civil/59341.
- Former member, Islamic Engineering Society, IIU Islamabad.

ADDITIONAL ACTIVITIES

- Selected participant, Cultural and Educational Exchange Program to the United States (HITEC College, 2013).

REFEREES

Available upon request. Reference letters will be submitted directly by referees through the IUSS online application system, in line with the ROSE PhD call procedure.